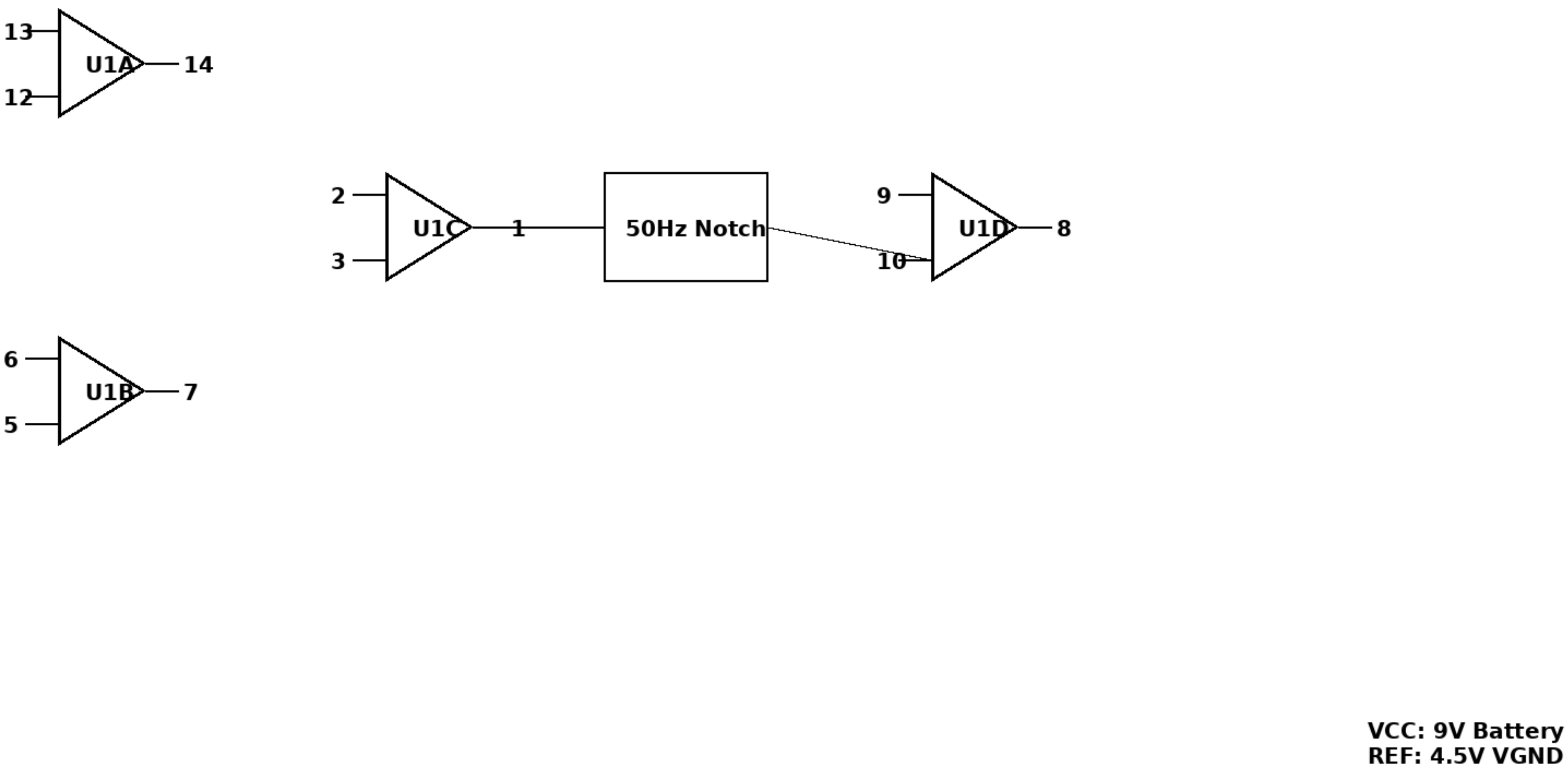
DIY EEG Circuit Guide (TL074 + Single 9V Battery)

Formal Circuit Schematic

EEG Circuit Schematic (Single TL074)



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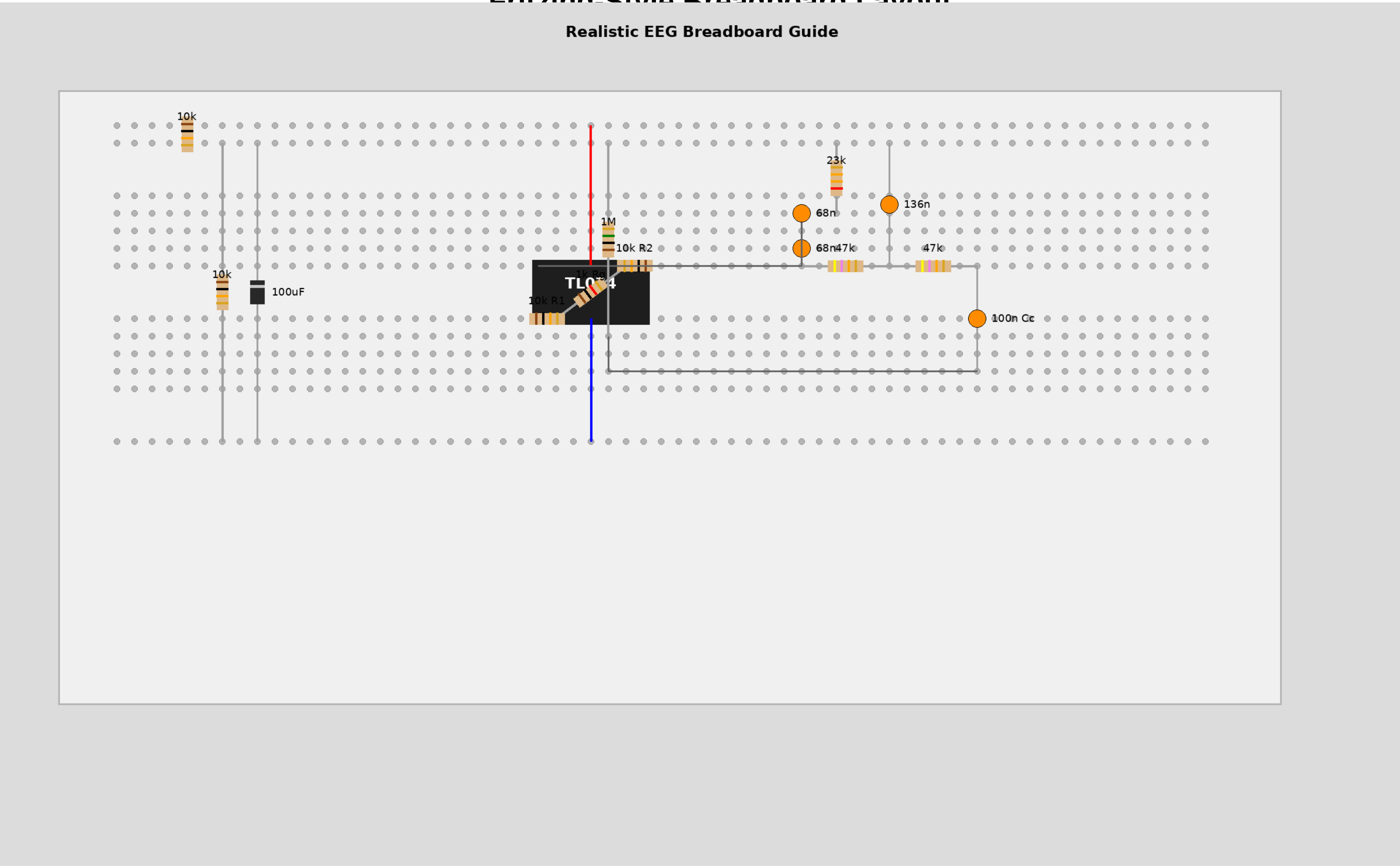
Fritzing-Style Breadboard Layout



DESIGN

EEG Circuit Design (Single TL074, Single 9V Battery)

Architecture

1. Virtual Ground (VGND):
   - Created using a resistor divider (2 \times 10k\Omega) and a 100\mu F capacitor.
   - VGND is \approx 4.5V. This acts as the "zero" for the analog signals.
2. Stage 1: Instrumentation Amplifier (InAmp)
   - Uses 3 op-amps.
   - Refers to VGND instead of Battery (-).
   - Gain \approx 21.
3. Stage 2: Passive Twin-T Notch Filter (50Hz)
   - Shunts to VGND.
4. Stage 3: Non-Inverting Active Bandpass Filter
   - Uses the 4th op-amp.
   - High Input Impedance (1M Ohm) referred to VGND.
   - Stage Gain \approx 471.
   - Total Gain \approx 9,890.

Power Supply

- Single 9V Battery.
- Battery (+): Connects to Pin 4 (VCC+).
- Battery (-): Connects to Pin 11 (VCC-) AND acts as the circuit's 0V return.
- Virtual Ground (VGND): 4.5V potential used as the reference for electrodes and filters.

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Schematic Description

TL074 Pinout (Top View)

1. Out 1 (InAmp Output)
2. In- 1 (InAmp Diff Stage)
3. In+ 1 (InAmp Diff Stage)
4. VCC+ (+9V)
5. In+ 2 (InAmp Input 2)
6. In- 2 (InAmp Input 2 - Internal)
7. Out 2 (InAmp First Stage Out 2)
8. Out 3 (Final Output)
9. In- 3 (Filter Stage)
10. In+ 3 (Filter Stage)
11. VCC- (0V / Battery Negative)
12. In+ 4 (InAmp Input 1)
13. In- 4 (InAmp Input 1 - Internal)
14. Out 4 (InAmp First Stage Out 1)

Virtual Ground Setup

- 10k Ohm from +9V to VGND rail.
- 10k Ohm from Battery (-) to VGND rail.
- 100uF Capacitor from VGND rail to Battery (-) [Observe Polarity: + to VGND].

Stage 1: Instrumentation Amplifier

- Inputs: Electrode A to Pin 12, Electrode B to Pin 5.

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- First Stage: 1k Ohm between Pin 13 and Pin 6. 10k Ohm from Pin 14 to 13. 10k Ohm from Pin 7 to 6.

- Diff Stage: 10k Ohm from Pin 14 to 2. 10k Ohm from Pin 7 to 3. 10k Ohm from Pin 3 to VGND. 10k Ohm from Pin 2 to Pin 1.

Stage 2: 50Hz Notch Filter

- Shunt components (Branch 1 capacitors and Branch 2 resistors) now connect to VGND.

Stage 3: Gain + Filter

- Pull-down resistor R_{pd} (1M Ohm) connects Pin 10 to VGND.

- Feedback shunt (1k Ohm + 47uF) connects to VGND.

Component List (Single Battery Version)

- 1x TL074CN Quad Op-Amp
- 1x 9V Battery
- Resistors (1%):
  - 2x 1k Ohm (R_g, R_{feedback_shunt})
  - 8x 10k Ohm (InAmp, VGND Divider)
  - 4x 47k Ohm (R_n1, R_n2 in upper branch, and 2x in parallel for 23.5k Ohm shunt)
  - 1x 470k Ohm (Final Gain)
  - 1x 1M Ohm (Input Pull-down)
- Capacitors:
  - 4x 68nF (Notch: 2x in series for branch, 2x in parallel for shunt)
  - 1x 10nF (Low Pass)
  - 1x 100nF (Coupling)
  - 1x 47uF (High Pass)

- 1x 100uF (VGND Stability)
- 2x 100nF (Decoupling)

ASSEMBLY

Breadboard Assembly - Single 9V Battery Version

Step 1: Virtual Ground (VGND) Setup

1. Battery (+) to Red Rail (+9V).
2. Battery (-) to Blue Rail (0V).
3. Place two 10k Ohm resistors in series from +9V rail to 0V rail.
4. The junction between the two resistors is your VGND. Wire this to a middle row or extra rail.
5. Place a 100uF capacitor between VGND and 0V rail (+ to VGND).

Step 2: IC Power

1. TL074 Pin 4 to +9V rail.
2. TL074 Pin 11 to 0V rail.

Step 3: Instrumentation Amplifier (Stage 1)

1. First Gain Stage: 1k Ohm (R_g) between Pin 13 and Pin 6.
2. 10k Ohm between Pin 14 and Pin 13.
3. 10k Ohm between Pin 7 and Pin 6.
4. Differential Stage: 10k Ohm from Pin 14 to Pin 2.
5. 10k Ohm from Pin 7 to Pin 3.
6. 10k Ohm from Pin 3 to VGND.
7. 10k Ohm from Pin 2 to Pin 1.

Step 4: 50Hz Notch Filter (Stage 2)

1. Upper Branch:

- Connect Pin 1 to a 47k Ohm resistor.
- Other end of resistor to a new row (Row X).
- From Row X, connect another 47k Ohm resistor to a new row (Row Y - Notch Output).
- From Row X, connect two 68nF capacitors in parallel to VGND.

2. Lower Branch:
   - Connect Pin 1 to a 68nF capacitor.
   - Other end of capacitor to a new row (Row Z).
   - From Row Z, connect another 68nF capacitor to Row Y.
   - From Row Z, connect two 47k Ohm resistors in parallel to VGND.

Step 5: High-Impedance Stage (Stage 3)

1. Place 100nF capacitor from Row Y (Notch Output) to Pin 10.
2. Place 1M Ohm resistor from Pin 10 to VGND.
3. Place 470k Ohm resistor and 10nF capacitor in parallel between Pin 8 and Pin 9.
4. Connect Pin 9 to a 1k Ohm resistor in series with a 47uF capacitor (+ side to resistor). The other side of the capacitor goes to VGND.

Step 6: Electrodes & Scope

1. Electrode A (Temple): Pin 12
2. Electrode B (Temple): Pin 5
3. Reference Electrode (Ear): VGND (Middle rail).
4. Oscilloscope:
   - Ground Clip to 0V rail (Battery negative).
   - Probe to Pin 8.
   - Settings: Set Oscilloscope to AC Coupling. Timebase: 200ms/div.

TESTING

Testing & Safety Procedures

WARNING: MANDATORY SAFETY RULES

FAILURE TO FOLLOW THESE RULES CAN RESULT IN ELECTRIC SHOCK.

1. SINGLE 9V BATTERY POWER ONLY: Never use a wall-plug power supply or more than one 9V battery in a way that bypasses the single-battery design.
2. OSCILLOSCOPE ISOLATION:
   - If using a mains-powered oscilloscope, ensure the circuit is only connected to the oscilloscope via the probe and ground clip.
   - Do not connect any other mains-powered devices (like a PC via USB) to the circuit while it is attached to a human.
   - Note: The oscilloscope ground is connected to Battery (-), while your body is connected to VGND (4.5V). This is safe as long as the battery is isolated from the mains.
3. NO LIQUID SPILLS: Keep Ten20 paste away from the breadboard. Keep all liquids away from the experimental setup.
4. DISCONNECT BEFORE MODIFYING: Always remove the electrodes from your body before making any changes to the circuit.

Step 1: Pre-Human Verification (Dry Run)

Before attaching electrodes to yourself, perform these tests:

1. Power Check:
   - Measure voltage between Pin 4 and Pin 11 (should be ~9V).

- Measure voltage between VGND and Pin 11 (should be ~4.5V).

2. Noise Floor Check:

- Short Pin 12 and Pin 5 to GND using jumper wires.

- Observe Pin 8 on the oscilloscope. You should see a very flat line with minimal noise (less than 100mV peak-to-peak).

3. Notch Filter Test (If possible):

- If you have a signal generator, inject a 50Hz, 100mV sine wave into Pin 12 (with Pin 5 at GND).

- The output at Pin 8 should be significantly attenuated at 50Hz compared to 40Hz or 60Hz.

Step 2: Human Testing (Alpha Wave Capture)

Alpha waves (8-13 Hz) are the easiest to see.

1. Electrode Placement:

- Reference: Earlobe or Mastoid (bone behind the ear).

- A & B: On the occipital lobe (back of the head) or the forehead.

2. Protocol:

- Sit in a comfortable chair and relax.

- Eyes Open: You should see chaotic, low-amplitude signals.

- Eyes Closed: After a few seconds of relaxation with eyes closed, Alpha waves (rhythmic oscillations) should appear on the oscilloscope.

- Eyes Open again: The Alpha waves should immediately disappear (Alpha blocking).

Troubleshooting

- Flat line at +9V or -9V: One of the op-amp stages is "saturated." Check for short circuits or missing resistors in the feedback loops.

- Heavy 50Hz Sine Wave: Your notch filter is not tuned or your grounding is poor. Check the Reference electrode connection and the capacitor values in the Twin-T filter.
- Random Spikes: Usually caused by moving your eyes, swallowing, or clenching your jaw (muscle artifacts). Stay as still as possible.